Simulation Report

Experiment 2 - Huffman Compression Accelerator on the PYNQ-Z2

1. Setup

Item	Value
Simulator	Icarus Verilog 12 (iverilog -g2012, SystemVerilog)
Design under test	huffman_top (RTL/*.sv)
Testbench	Testbench/huffman_tb.sv with the code table Testbench/huffman_codes.svh
Waveform dump	Simulation/wave_dump.sv (records the main signals to huffman_wave.vcd)
Clock	100 MHz (10 ns period), same as the board
Input message	"hello world, huffman on a pynq!!" (32 bytes, 256 bits)
Mode	SW0 = 0, so the engine compresses

2. What the testbench checks

The testbench drives the AXI4-Lite port of **huffman_top** the same way the ARM processor does on the board, with two tasks *axi_write* and *axi_read*. While it sends the message it also builds the expected output itself: it looks up every character in the same code table the hardware uses (huffman_codes.svh, generated by the same script as the ROM) and packs the bits into 32-bit words in the same order.

#	Step	Expected result
1	Write 1 to CTRL (0x00)	Engine and counters are reset
2	Write the 32 characters to DATA IN (0x08), one per write	Each character is coded and packed
3	Write 2 to CTRL (flush)	The last partial word is sent, padded with zeros
4	Read BYTES IN (0x10)	32
5	Read BITS OUT (0x14)	Sum of the code lengths = 179
6	Read STATUS (0x04)	6 words waiting in the FIFO
7	Read DATA OUT (0x0C) six times	Every word equals the expected packed word

Any mismatch stops the simulation with *$fatal*, so the PASS line means every check passed.

3. Result

PASS - 32 bytes (256 bits) were compressed into 179 bits in 6 words, and all six words were exactly the expected ones. That is 5.6 bits per character, 70 % of the original size, for this short message with many rare characters (",", "!", "q").

The "sorry: constant selects" line is a known Icarus Verilog limitation message about the always_comb block in huffman_axi_wrapper.sv. It does not change the result.

Transcript

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Experiment 2 - Huffman Compression Accelerator (PYNQ-Z2)
Simulation transcript, 2026-09-18 15:30
Simulator: Icarus Verilog version 12.0 (devel) (s20150603-1539-g2693dd32b)

> iverilog -g2012 -I ../Testbench -o huffman_sim.vvp -s huffman_tb -s wave_dump ../RTL/*.sv ../Testbench/huf
fman_tb.sv wave_dump.sv   (run inside Simulation/)
../RTL/huffman_axi_wrapper.sv:143: sorry: constant selects in always_* processes are not currently supported
(all bits will be included).

> vvp huffman_sim.vvp
VCD info: dumpfile huffman_wave.vcd opened for output.
bytes in  = 32 (expected 32)
bits out  = 179 (expected 179)
status    = 00060002, 6 words waiting
compressed 32 bytes (179 bits) into 6 words, all correct
compression ratio: 179 bits out for 256 bits in
PASS: all Huffman accelerator checks passed
../Testbench/huffman_tb.sv:151: $finish called at 2785000 (1ps)
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4. Waveform

(a) The first four characters are written to 0x08. One clock after *char_valid* the ROM gives the code and its length ("h" = 5 bits, "e" = 3, "l" = 5); the next clock the code is shifted into *bit_buffer* above the bits already there and *bit_count* grows by the code length. (b) When *bit_count* would pass 32 the low 32 bits leave as a packed word (0x2E39AD95), *valid_out* pulses and the word enters the FIFO; the 3 bits left over stay in the buffer. (c) A flush sends the 19 bits still in the buffer as the sixth word and pulses *done*. (d) Software reads the counters (0x20 = 32 bytes, 0xB3 = 179 bits), the status (6 words) and then the six words, and the FIFO count drops to zero.

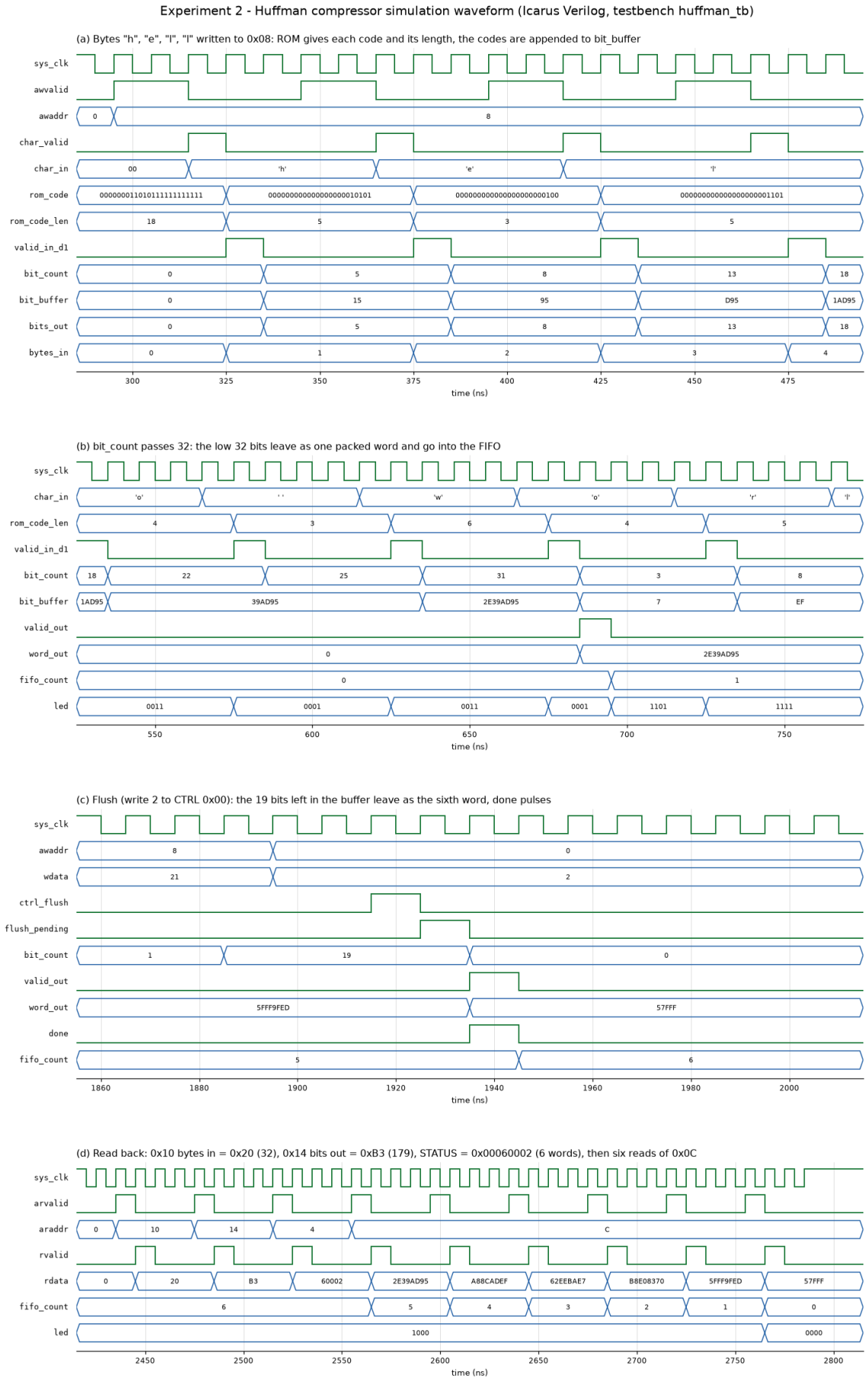


Figure 1 - Simulation waveform of huffman_tb (drawn from huffman_wave.vcd)